

Yash Siddhapura

Electronics & Communication Engineering

L.D. College Of Engineering

siddhapurayash09@gmail.com | +91 7284985820 | [in Yash Siddhapura](https://www.linkedin.com/in/Yash-Siddhapura) | [Siddhapura-Yash](https://github.com/Siddhapura-Yash)

EDUCATION

- B.E. in Electronics & Communication** 2023-26
L.D. College of Engineering, Ahmedabad
CGPA : 7.99/10
- Diploma in Electronics & Communication** 2020-23
A.V. Parekh Technical Institute, Rajkot
CGPA : 9.28/10

WORK EXPERIENCE & INTERNSHIPS

- FPGA Developer Intern | Vicharak** (Jan'26 - present)
 - Implemented and verified UART, I2C, and UART-to-DDR data transfer pipelines in Verilog on FPGA hardware (Vaaman board).
 - Replaced vendor FIFO IP with custom synchronous/asynchronous FIFOs using Gray-code CDC, achieving 151 MHz operation with positive timing slack across 40–120 MHz clock domains.
 - Independently designed an FPGA-based stereo vision accelerator by studying research papers and existing stereo-matching architectures for real-time depth estimation.
 - Developed streaming and pipelined RTL architectures for Census Transform, Absolute Difference (AD), cost aggregation, and SGM-inspired disparity computation modules.
 - Optimized the accelerator for high-throughput execution through line-buffer-based memory reuse, BRAM/bandwidth analysis, and reduced DDR access for real-time image processing.
- Intern | Power Electronics** (July'23 - Sep'23)
 - Worked on transformer winding and testing, gaining hands-on experience in power electronics.
 - Developed skills in electrical component analysis and circuit testing.

PROJECTS

- 32-bit ARM-like RISC Processor in Verilog**
 - Designed and implemented a 32-bit ARM-like RISC CPU in Verilog using modular RTL design.
 - Developed core components including ALU, register file, datapath, control unit, instruction decoder, and memory interface.
 - Supported arithmetic, logical, and memory instructions (ADD, SUB, MOV, CMP, B, LDR, STR) with verified fetch-decode-execute operation on FPGA (Vaaman board).
- SHA-256 Cryptographic Hash Function in Verilog**
 - Implemented SHA-256 within a single Verilog module using functions for logical and arithmetic operations.
 - Performed message preprocessing, scheduling, and 64-round compression in a compact RTL design.
 - Demonstrated understanding of cryptographic operations and hardware-based hash computation.
- VGA Signal Generator in Verilog**
 - Designed VGA timing controller generating HSYNC, VSYNC, and pixel timing signals.
 - Implemented horizontal and vertical counters for frame synchronization.
 - Verified display timing and signal correctness through simulation.

TECHNICAL SKILLS

- Programming Languages** : Verilog, SystemVerilog (Basics), VHDL, C, C++, Python, Linux (Essential Commands)
- Simulation & EDA Tools**: Modelsim, Quartus, Vivado, EDA Playground, iverilog, Effinity, Proteus

ACHIEVEMENTS & AWARDS

- Awarded a ₹30,000 SSIP Grant for the Master-Slave Robotic Arm project.
- Achieved Rank 50 (Gujarat, all branches) in B.Tech Lateral Entry admissions, 2023.
- Strengthened RTL design and logic fundamentals by solving 141+ verilog challenges on HDLBits.